A Weighted Prefix–Tail Gluing Criterion for Regularized Determinants

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Abstract

The new spectral-tail certificate and the deterministic numerator envelope live on opposite sides of a missing prefix interface. We formulate the exact gluing theorem. Split the logarithmic coefficient difference at a moving order m_σ. If the weighted prefix mismatch, the noisy high-order tail, and the target high-order tail all vanish on a closed disk, then the normalized determinant quotient converges uniformly there, with an explicit exponential error bound. RH-285 supplies the noisy spectral-tail leaf when the noisy coefficient sequence is the modulus-complement trace, and the deterministic envelope papers supply the target leaf. The remaining prefix has a typed two-error decomposition: total noisy trace versus counterloop plus anchor, and noisy head versus counterloop. RH-287 controls only the first error unweighted and without a rate. The criterion prevents finite-order agreement from being promoted to a determinant theorem.

1 Three budgets

Let

$$F_\sigma(z) = \exp\left(-\sum_{n\geq 2} \frac{b_{\sigma,n}}{n} z^n\right), \qquad F(z) = \exp\left(-\sum_{n\geq 2} \frac{a_n}{n} z^n\right)$$

be normalized logarithmic germs whose displayed logarithmic series converge absolutely and uniformly on the closed disk $|z| \leq R$. For an integer $m_\sigma \geq 3$, put

$$\begin{aligned} P_\sigma(R) &= \sum_{2\leq n<m_\sigma} \frac{|b_{\sigma,n} - a_n|R^n}{n}, \\ S_\sigma(R) &= \sum_{n\geq m_\sigma} \frac{|b_{\sigma,n}|R^n}{n}, \\ T_\sigma(R) &= \sum_{n\geq m_\sigma} \frac{|a_n|R^n}{n}. \end{aligned}$$

2 Gluing theorem

Theorem 2.1. *If $P_\sigma(R) + S_\sigma(R) + T_\sigma(R) \to 0$, then*

$$\sup_{|z|\leq R} \left| \frac{F_\sigma(z)}{F(z)} - 1 \right| \leq \exp\{P_\sigma(R) + S_\sigma(R) + T_\sigma(R)\} - 1 \longrightarrow 0.$$

Proof. On $|z| \leq R$, split the normalized logarithm

$$\log \frac{F_\sigma(z)}{F(z)} = -\sum_{2\leq n<m_\sigma} \frac{(b_{\sigma,n} - a_n)z^n}{n} - \sum_{n\geq m_\sigma} \frac{b_{\sigma,n}z^n}{n} + \sum_{n\geq m_\sigma} \frac{a_n z^n}{n}.$$

Its modulus is at most $P_\sigma + S_\sigma + T_\sigma$. The inequality $|e^w - 1| \leq e^{|w|} - 1$ gives the result. $\square$

3 Typed spectral application

Let $c_{\sigma,n}$ be the total noisy bulk trace, let

$$h_{\sigma,n} = \sum_{\mu \in H_\sigma} \mu^n, \qquad \tau_{\sigma,n} = c_{\sigma,n} - h_{\sigma,n} = \operatorname{Tr} C_\sigma^n,$$

and let $s_{k_\sigma,n}$ be the finite-radius counterloop moment. The target anchor is a_n. Define the two typed errors

$$e_{\sigma,n} = c_{\sigma,n} - s_{k_\sigma,n} - a_n, \qquad d_{\sigma,n} = h_{\sigma,n} - s_{k_\sigma,n}.$$

Then the coefficient needed for the spectral complement satisfies the exact identity

$$\tau_{\sigma,n} - a_n = e_{\sigma,n} - d_{\sigma,n}. \tag{1}$$

Consequently its prefix budget obeys

$$P_\sigma(R) \le E_\sigma(R) + D_\sigma(R),$$

where

$$E_\sigma(R) = \sum_{2 \le n < m_\sigma} \frac{|e_{\sigma,n}| R^n}{n}, \qquad D_\sigma(R) = \sum_{2 \le n < m_\sigma} \frac{|d_{\sigma,n}| R^n}{n}.$$

Thus one may prove the missing prefix either directly for $\tau_{\sigma,n} - a_n$, or by proving both typed weighted estimates. Head-to-counterloop transport alone does not imply the complement-to-anchor estimate.

4 Current leaf audit

In the gluing theorem set $b_{\sigma,n} = \tau_{\sigma,n}$. The target trace envelope and coefficient-radius law give $T_\sigma(R) \to 0$ on the certified strict target disk. RH-282 and RH-285 give $S_\sigma(R) \to 0$ for this projection-free modulus complement after the logarithmic block clock. The unresolved term is the direct complement-to-anchor prefix $P_\sigma(R)$.

RH-287 proves an unweighted maximum for $e_{\sigma,n}$ on some growing prefix, but its only automatic weighted consequence is

$$E_\sigma^{(h)}(R) := \sum_{2 \le n \le h_\sigma} \frac{|e_{\sigma,n}| R^n}{n} \le \varepsilon_\sigma \sum_{2 \le n \le h_\sigma} \frac{R^n}{n}.$$

The clock h_σ need not reach the spectral tail clock m_σ, and for $R > 1$ even the displayed partial bound need not vanish. No current theorem supplies the weighted $D_\sigma(R)$ term either. A direct $P_\sigma \to 0$ theorem, or both typed weighted estimates in (1), is still required.

Remark 4.1. The theorem is an assembly criterion, not a claim that its three leaves are all present. Gate A and Gates B–E remain false/open. It does not construct a Hilbert–Polya operator or identify any arithmetic divisor.